

Basics

An electromagnetic field is a physical field produced by electrically charged particles.

This field is a combination of an electric field and magnetic field.

Electric field is produced by stationary charges and magnetic field by moving charges.

Scalar qty:- Qty characterised by magnitude is called scalar qty - ex: mass, time, temp etc.

vector qty:- Both magnitude & direction - Force, velocity, acceleration.

Scalar & vector field

Field:- Region in which a particular physical function has a value at each & every pt in the region.

Scalar field:- If scalar qty is specified in a region - scalar field.

Atmospheric temp & pressure in a region.
Electrostatic potential.

Vectors field:-

Vectors qty is specified in a region, then region is said to be vector field.

ex: gravitational force & wind velocity in atmosphere
magnetic field produced by a current carrying conductor

vector

Unit vector - $\text{vector} = \frac{\vec{OA}}{|\vec{OA}|}$
Magnitude $|\vec{OA}|$

Vector multiplication:-

1) scalar mptn / dot product

Scalar mptn of 2 vectors is a scalar qty whose magnitude is the product of magnitudes of vectors and cosine of angle b/w them.

product of
 $\vec{A} \cdot \vec{B}$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \alpha$$

$$= A_x A_x + A_y A_y + A_z A_z$$

$$= B_x A_x + B_y A_y + B_z A_z$$

$$\vec{B} = [B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z] \cdot [A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z]$$

$$= B_x A_x + A_y B_y + A_z B_z$$

$$A_x \cdot A_x = A_y \cdot A_y = A_z \cdot A_z = 1$$

$$A_x \cdot A_y = A_y \cdot A_z = A_z \cdot A_x = 0$$

2) vector product / cross product

It is the product of magnitudes of vectors & sine of angle b/w them.

The product will be a vector & its direction is

perp to plane containing the 2 vectors.

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \alpha$$

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z \quad \& \quad \vec{B} = B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

NOTE: $\Rightarrow$ If $\vec{A} \cdot \vec{B} = 0$ - vectors are orthogonal

$\vec{A} \cdot \vec{B}$ are perpendicular

If $\vec{A} \times \vec{B} = 0$, then vectors are collinear & parallel

Q1)

If $\vec{A} = 2a_x + 2a_y - a_z$ and $\vec{B} = 6a_x - 3a_y + 2a_z$

1) Find $\vec{A} \cdot \vec{B}$

$$\vec{A} \cdot \vec{B} = 12 - 6 - 2 = 4$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 2 & 2 & -1 \\ 6 & -3 & 2 \end{vmatrix} = a_x [4 - 3] - a_y [4 + 6] + a_z [-6 - 12]$$

$$A \cdot B = 4a_x + 3a_y + a_z \quad \& \cdot B = 2a_x - a_y + 2a_z$$

find a) a unit vector $\perp$ to A & B

b) Angle b/w A & B

$$\begin{aligned} \vec{A} \times \vec{B} &= \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 4 & 3 & 1 \\ 2 & -1 & 2 \end{vmatrix} = \hat{a}_x [6+1] - \hat{a}_y [8-2] \\ &\quad + \hat{a}_z [-4-6] \\ &= 7\hat{a}_x - 6\hat{a}_y - 10\hat{a}_z \end{aligned}$$

$$\text{unit vector } \perp \text{ to } A \& B = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \frac{7\hat{a}_x - 6\hat{a}_y - 10\hat{a}_z}{\sqrt{49+36+100}}$$

$$= \frac{7\hat{a}_x - 6\hat{a}_y - 10\hat{a}_z}{\sqrt{185}}$$

$$b) \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$= 8 - 3 + 2 = 7$$

$$A \cdot B = |\vec{A}| |\vec{B}| \cos \theta$$

$$|\vec{A}| = \sqrt{16+9+1} = \sqrt{26}$$

$$|\vec{B}| = \sqrt{4+1+4} = \sqrt{9}$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \frac{7}{3\sqrt{26}} = 0.45$$

S. $\vec{T}$ vectors $\vec{A} = 6a_x + 2a_y - 5a_z$ & $\vec{B} = 5a_x - 5a_y + 4a_z$ are $\perp$ to each other.

$$A \cdot B = 0$$

$$\vec{A} \cdot \vec{B} = 30 - 10 - 20 = 0$$

2) S. $\vec{T}$ vectors $\vec{A} = 4a_x - 2a_y + 2a_z$ & $\vec{B} = -6a_x + 3a_y - 3a_z$ are

$\perp$ to each other.

$$\vec{A} \cdot \vec{B} = 0$$

$$\vec{A} \times \vec{B} =$$

$$\begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 4 & -2 & 2 \\ -6 & 3 & -3 \end{vmatrix}$$

$$= \hat{a}_x [6-6] - \hat{a}_y [12+12] + \hat{a}_z [12-12] = 0$$

vectors are $\perp$.

Coordinate system :-

It is a s/m which is used to represent a pt in space.

3 type of coordinate s/m are

1) Rectangular / Cartesian s/m.

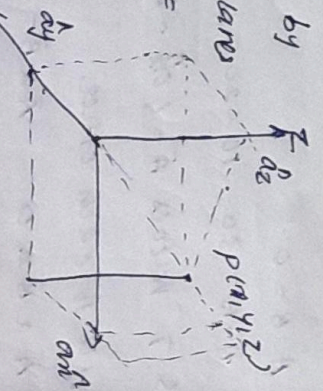
2) Cylindrical coordinate s/m

3) Spherical coordinate s/m.

Rectangular / Cartesian co-ordinate s/m.

In this s/m 3 co-ordinate axes x, y, z are mutually $\perp$ to each other, 3 axes intersect at a common pt called origin of the s/m.

A point P is defined by the integration of 3 planes
 $x = \text{const}$, $y = \text{const}$, $z = \text{const}$
 (out planes).



variation of x, y, z
 axis

$$-\infty \leq x \leq \infty$$

$$-\infty \leq y \leq \infty$$

$$-\infty \leq z \leq \infty$$

Position vector of point P

$$\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$$

$$\text{magnitude } |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

Differential elements

consider a pt P(x, y, z) in rectangular co-ordinates. In

lets increase each coordinate by a differential amt.

A point P' is obtained with coordinates (x+dx, y+dy, z+dz)

dx - differential length in x direction

dy - differential length in y direction

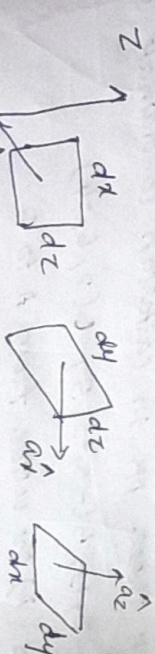
dz - differential length in z direction

Differential length

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

$$|d\vec{l}| = \sqrt{dx^2 + dy^2 + dz^2}$$

Differential surface area :-



$$d\vec{s}_x = dydz\hat{a}_x$$

$$d\vec{s}_y = dx dz\hat{a}_y$$

$$d\vec{s}_z = dx dy\hat{a}_z$$

Diff. Vol. :-

$$dV = dx dy dz$$

Diff. Volume is a rectangular parallelepiped.

1.2 Two pts A(2, 2, 1) & B(3, -4, 2) are given in

the cartesian system. Obtain the vector from A to B & a unit vector directed from A to B.

$$\vec{A} = 2\hat{a}_x + 2\hat{a}_y + \hat{a}_z \quad \& \quad \vec{B} = 3\hat{a}_x - 4\hat{a}_y + 2\hat{a}_z$$

$$\vec{AB} = \vec{B} - \vec{A} = (3-2)\hat{a}_x + (-4-2)\hat{a}_y + (2-1)\hat{a}_z$$

$$\vec{AB} = \hat{a}_x - 6\hat{a}_y + \hat{a}_z$$

Unit vector

$$\vec{a}_{AB} = \frac{\vec{AB}}{|\vec{AB}|} = \frac{a_x\hat{i} - 6a_y\hat{j} + a_z\hat{k}}{6.164} = 0.16a_x - 0.97a_y + 0.162a_z$$

$$|\vec{a}_{AB}| = 1$$

Given three pts in cartesian co-ordinate system

$$A(3, -2, 1), B(-3, -3, 5), C(2, 6, -4)$$

And i) The vector from A to C

ii) The unit vector from A to C

iii) The distance from A to C

iv) The vector from A to the midpoint of the segment joining B to C.

Position vectors are

$$\vec{A} = 3a_x\hat{i} - 2a_y\hat{j} + a_z\hat{k}, \quad \vec{B} = -3a_x\hat{i} - 3a_y\hat{j} + 5a_z\hat{k}$$

$$C = 2a_x\hat{i} + 6a_y\hat{j} - 4a_z\hat{k}$$

i) Vector from A to C

$$\vec{AC} = \vec{C} - \vec{A} = -6a_x\hat{i} + 8a_y\hat{j} - 5a_z\hat{k}$$

$$\vec{BA} = \vec{A} - \vec{B} = 6a_x\hat{i} + a_y\hat{j} - 4a_z\hat{k}$$

$$|\vec{BA}| = \sqrt{6^2 + 1^2 + 16} = 7.28$$

$$\vec{a}_{BA} = \frac{\vec{BA}}{|\vec{BA}|} = 0.82a_x + 0.13a_y - 0.54a_z$$

ii) Distance b/w B & C

$$\vec{BC} = \vec{C} - \vec{B} = 5a_x + 9a_y - 9a_z$$

$$|\vec{BC}| = \sqrt{5^2 + 9^2 + 81} = 13.61$$

$$\text{Distance} = \sqrt{5^2 + 9^2 + 81} = 13.61$$

$$iv) B(x_1, y_1, z_1) \& C(x_2, y_2, z_2)$$

$$\text{midpoint of BC} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

$$\text{Midpoint of BC} = \left(\frac{-3+2}{2}, \frac{-3+6}{2}, \frac{5-4}{2} \right)$$

$$= (-0.5, 1.5, 0.5)$$

vector from A to this mid pt

$$= [-0.5 - 3]a_x + [1.5 + 2]a_y + [0.5 - 1]a_z$$

$$= -3.5a_x + 3.5a_y - 0.5a_z$$

Cylindrical co-ordinate sys.

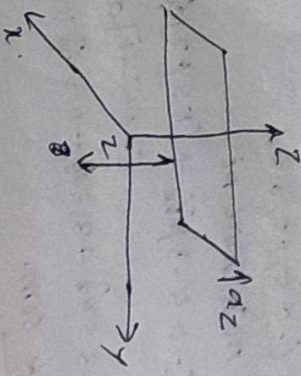
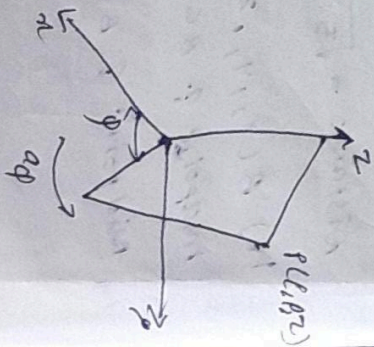
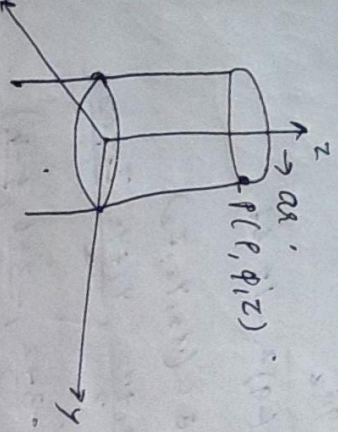
Consider a pt P in the cylindrical coordinate sys which is the intersection of 3 mutually perpendicular surfaces. They are:

i) cylinder of radius 'r' with z axis as the axis of cylinder

ii) a half plane perpendicular to the x-y plane and at an angle 'φ' with the x-z plane.

iii) another plane with const 'z' which is vertical

to x - y plane.

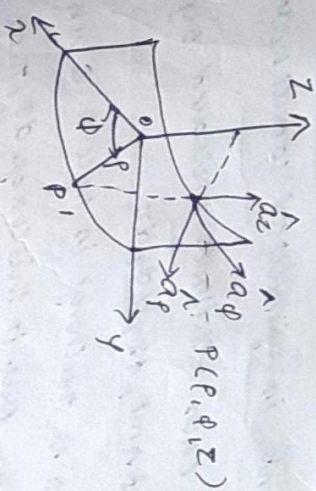
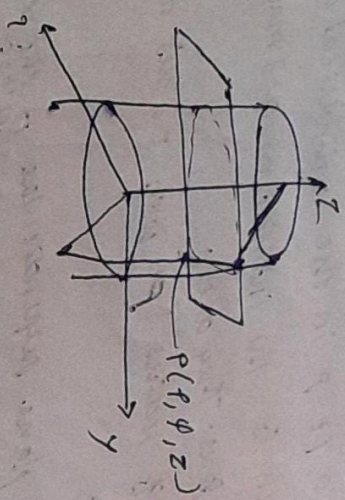


The ranges of the variables are

$$0 \leq r \leq \infty$$

$$0 \leq \phi \leq 2\pi$$

$$-\infty \leq z \leq \infty$$

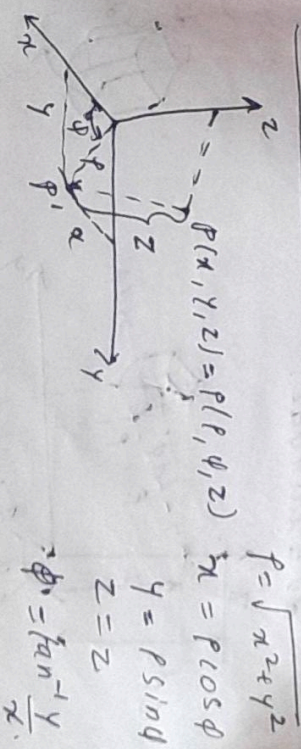


$\vec{a}_r$ lies in a plane $\vec{a}_\phi$ lies parallel to xy plane but it is tangent to cylinder & pointing in a direction of ϕ inc. $\vec{a}_z$ is del to z axis and directed towards increasing z

Vector $\vec{r}$ can be represented as

$$\vec{r} = r \vec{a}_r + r \sin \phi \vec{a}_\phi + z \vec{a}_z$$

Relation b/w cylindrical & cartesian sys.



$$r = \sqrt{x^2 + y^2}$$

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

$$\phi = \tan^{-1} \frac{y}{x}$$

Similar to cartesian coordinate sys

These are 3 unit vectors in the r, ϕ, z directions denoted as $\vec{a}_r, \vec{a}_\phi$ and $\vec{a}_z$

$\vec{a}_z$ is $\uparrow$ to surface

$\vec{a}_r$ is $\uparrow$ to surface

$\vec{a}_\phi$ is $\uparrow$ to surface

Differential elements in cylindrical coordinate system

$P(r, \phi, z)$ - let each coordinate increase by differential element, $dr, d\phi, dz$

In ϕ direction, $d\phi$ is the change in angle ϕ & is not the differential length. Due to this change $d\phi$, there will be a differential arc length in ϕ direction. This arc length is $r d\phi$.

ds - Differential length in r direction

$rd\phi$ - " " " ϕ " "

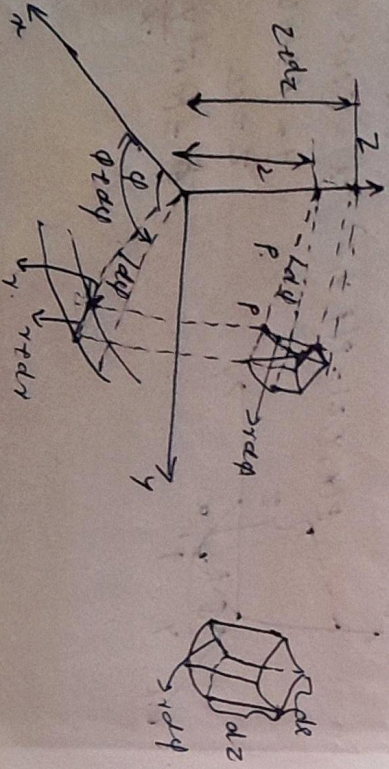
dz - " " " z " "

$$dl = ds \bar{e}_r + r d\phi \bar{e}_\phi + dz \bar{e}_z$$

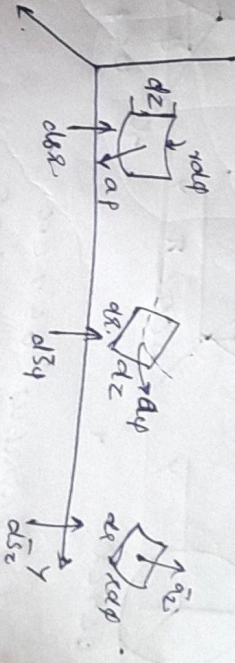
The magnitude of diff. length = $\sqrt{(ds)^2 + (r d\phi)^2 + (dz)^2}$

Differential volume

$$dV = ds dr d\phi dz$$



Differential Surface areas



$$ds_{\bar{e}_r} = \text{diff. vector surface area normal to } r \text{ direction} = r d\phi dz \bar{e}_r$$

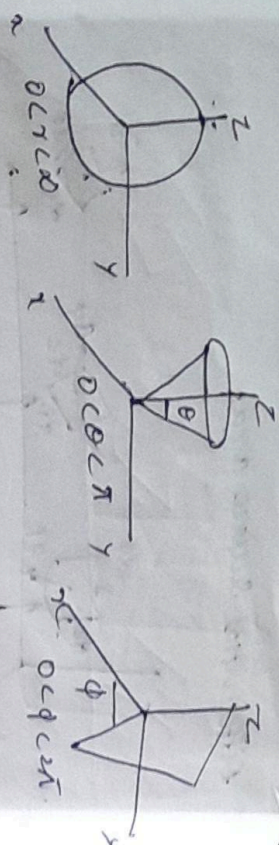
$$ds_{\bar{e}_\phi} = \text{diff. vector surface area normal to } \phi \text{ direction} = dr dz \bar{e}_\phi$$

$$ds_{\bar{e}_z} = \text{diff. vector surface area normal to } z \text{ direction} = r dr d\phi \bar{e}_z$$

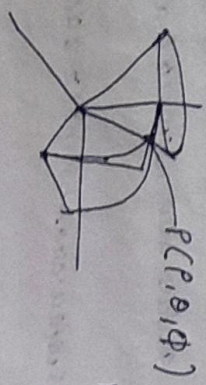
Spherical co-ordinate system

Surfaces used to define the spherical coordinate system are

- 1) Sphere of radius r , origin as centre of sphere
 - 2) Right circular cone with apex at origin & its axis as z axis. Its half angle is θ . It rotates about z axis and θ varies from 0 to π .
 - 3) A half plane ϕ to 2π plane.
- any making an angle ϕ with zx plane.



Ranges of variables
 $0 \leq r < \infty$
 $0 \leq \theta \leq 2\pi$
 $0 \leq \phi \leq \pi$



Vector
 $\vec{P} = r \hat{a}_r + r \sin \theta \hat{a}_\theta + r \cos \theta \hat{a}_\phi$

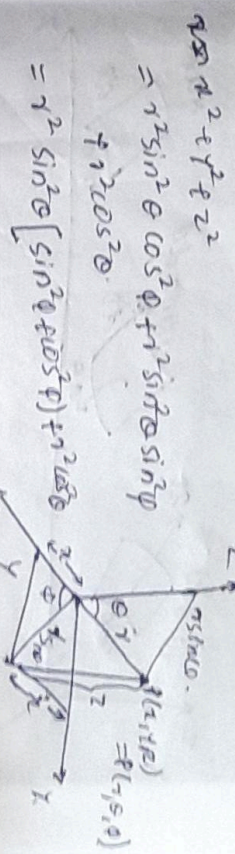
$\hat{a}_r$ - unit vector directed from centre of sphere
 Normally extrinsals

$\hat{a}_\theta$ - unit vector tangentially to the sphere & directed in the direction of θ

$\hat{a}_\phi$ - unit vector tangentially to both the sphere and conical surface & directed in the direction of ϕ

Relation b/w Cartesian & spherical Sins

$x = r \sin \theta \cos \phi$ $y = r \sin \theta \sin \phi$ $z = r \cos \theta$



$= r^2 (1)$

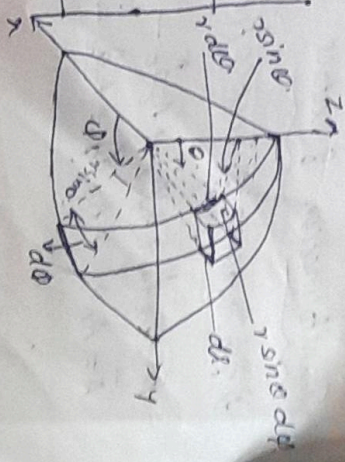
$r = \sqrt{x^2 + y^2 + z^2}$
 $\tan \phi = y/x$ $\cos \theta = z/r$

while using the formulas care must be taken to place the angles $\theta \leq \phi$ in the correct quadrant acc to signs of x, y, z

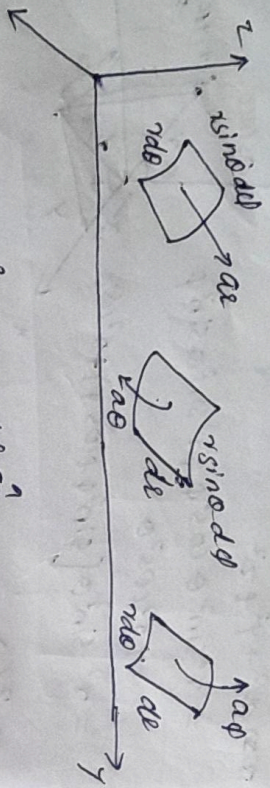
diff. elements in spherical co-ordinates
 $\rho(r, \theta, \phi)$

differentiated length:
 $dl = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin \theta d\phi \hat{a}_\phi$

Differential elements in coordinates	Corresponding change in lengths
dr	dr
$d\theta$	$r d\theta$
$d\phi$	$r \sin \theta d\phi$



* diff. vol.
 $dV = dr \times r d\theta \times r \sin \theta d\phi$
 $= r^2 \sin \theta dr d\theta d\phi$



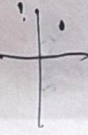
$$dV = r^2 \sin \theta dr d\theta d\phi$$

$$dV = r^2 \sin \theta dr d\theta d\phi$$

$$dV = r^2 \sin \theta dr d\theta d\phi$$

Given pt $P(-2, 6, 3)$. Express in spherical co-ordin.

At sim.



$$x = -2, y = 6, z = 3$$

$$r = \sqrt{4 + 36 + 9} = 7$$

$$\theta = \cos^{-1} \frac{z}{r} = \frac{3}{7} = 64.2^\circ$$

$$\phi = \tan^{-1} y/x = \tan^{-1} -3 =$$

Transformation of vectors

1) Transformation of vectors from cartesian to cylindrical

$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$ in cartesian co-ordination.

$\vec{A} = A_r \hat{a}_r + A_\phi \hat{a}_\phi + A_z \hat{a}_z$ in cylindrical co-ordination.

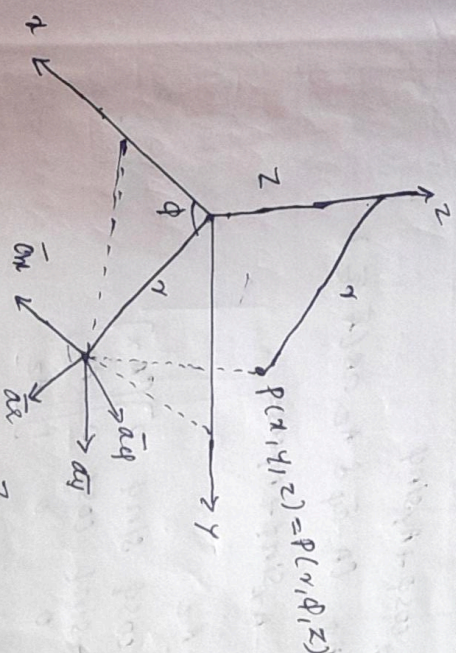
Component of vector in the direction of any unit vector is dot product with that unit vector. Hence component of $\vec{A}$ in the direction of $\hat{a}_r$ is $\vec{A} \cdot \hat{a}_r$.

$$A_r = (\vec{A} \cdot \hat{a}_r)$$

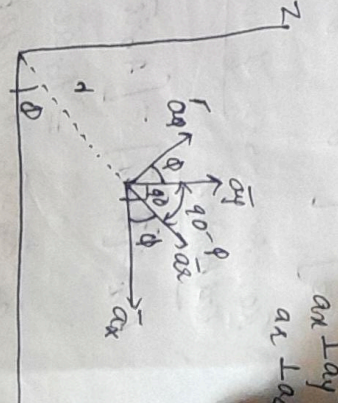
$$= [A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z] \cdot \hat{a}_r$$

$$= A_x \hat{a}_r \cdot \hat{a}_r + A_y \hat{a}_y \cdot \hat{a}_r + A_z \hat{a}_z \cdot \hat{a}_r$$

magnitude of all unit vectors is unity, hence it is necessary to find angle b/w unit vectors to obtain various dot products.



- $\angle$ b/w $\hat{a}_r$ & $\hat{a}_z$ is θ
- $\angle$ b/w $\hat{a}_r$ & $\hat{a}_x$ is $90^\circ - \theta$
- $\angle$ b/w $\hat{a}_r$ & $\hat{a}_y$ is $90^\circ + \theta$
- $\angle$ b/w $\hat{a}_\phi$ & $\hat{a}_x$ is ϕ
- $\angle$ b/w $\hat{a}_\phi$ & $\hat{a}_y$ is $90^\circ - \phi$
- $\angle$ b/w $\hat{a}_\phi$ & $\hat{a}_z$ is $90^\circ + \phi$



Σ/A
π/2

$$\bar{a}_x \cdot \bar{a}_x = \cos \phi$$

$$\bar{a}_x \cdot \bar{a}_y = \cos (90^\circ + \phi)$$

$$= -\sin \phi$$

$$\bar{a}_y \cdot \bar{a}_x = \cos (90^\circ - \phi)$$

$$= \sin \phi$$

Staircase
cos
tan
u.s

$$\bar{a}_y \cdot \bar{a}_y = \cos \phi$$

$\bar{a}_z \cdot \bar{a}_z = \bar{a}_z \cdot \bar{a}_y = 0$ as $\bar{a}_z$ is perpendicular to $\bar{a}_x$ & $\bar{a}_y$

$$\bar{a}_z \cdot \bar{a}_z = 1$$

$$A_x = A_x \cos \phi + A_y \sin \phi$$

$$\text{Similarly } A_y = A_x \sin \phi + A_z \cos \phi$$

$$A_z = -A_x \sin \phi + A_y \cos \phi$$

$$A_z = A_z$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Transformation of vector from cylindrical to Cartesian

$$\bar{A}_z = [\bar{A} \cdot \bar{a}_z] = [A_x \bar{a}_x + A_y \bar{a}_y + A_z \bar{a}_z] \cdot \bar{a}_z$$

$$= A_x \bar{a}_x \cdot \bar{a}_z + A_y \bar{a}_y \cdot \bar{a}_z + A_z \bar{a}_z \cdot \bar{a}_z$$

$$\bar{a}_x \cdot \bar{a}_z = \bar{a}_y \cdot \bar{a}_z = 0 \text{ (dot product is commutative)}$$

$$= \cos \phi$$

$$A_x = A_x \cos \phi - A_y \sin \phi$$

$$A_y = A_x \sin \phi + A_z \cos \phi$$

$$A_z = A_z$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Dot product	$\bar{a}_x$	$\bar{a}_y$	$\bar{a}_z$
$\bar{a}_x$	$\cos \phi$	$-\sin \phi$	0
$\bar{a}_y$	$\sin \phi$	$\cos \phi$	0
$\bar{a}_z$	0	0	1

Transform the vector field $\vec{w} = 10\bar{a}_x - 8\bar{a}_y + 6\bar{a}_z$ to cylindrical coordinate system at pt $P(10, 8, 4)$

$$w_x = 10 \quad w_y = -8 \quad w_z = 6$$

$$w_x = \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$$

$$\phi = \tan^{-1} y/x$$

$$= \tan^{-1} \left[\frac{-8}{10} \right]$$

$$= -38.65^\circ$$

$$\phi \rightarrow 4th \text{ quadrant}$$

$$\phi = 360 - 38.65 = 321.35^\circ$$

$$P = \sqrt{x^2 + y^2} = \sqrt{10^2 + 8^2} = 12.8$$

$$w_p =$$

$$w_p = w_x \cos \phi + w_y \sin \phi + 0$$

$$= 10 \cos 321.34 - 8 \sin 321.34$$

$$= 14.80$$

$$w_\phi = -w_x \sin \phi + w_y \cos \phi + 0$$

$$= -10 \sin 321.34 - 8 \cos 321.34 =$$

$$= 0$$

$$w_z = 6$$

$$\vec{w} = 14.801 \hat{a}_p + 6 \hat{a}_z$$

Given pt $(-2, 6, 3)$ and vector $\vec{A} = y\hat{a}_x + (x+z)\hat{a}_y$.
Evaluate $\vec{A}$ at pt P in cylindrical coordinate system.

$$\begin{bmatrix} A_p \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\phi = \tan^{-1}(y/x) = -71.56^\circ$$

$$\phi = 180 + (-71.56) = 108.44$$

$$\vec{A} = y\hat{a}_x + (x+z)\hat{a}_y + 0$$

$$A_x = y \quad A_y = x+z \quad A_z = 0$$

$$A_p = A_x \cos \phi + A_y \sin \phi$$

$$= y \cos 108.44 + (x+z) \sin 108.44$$

$$= -0.945$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$= -y \sin \phi + (x+z) \cos \phi$$

$$= -5.7$$

$$A_z = 0$$

$$A_p = -0.94 \hat{a}_p - 5.7 \hat{a}_\phi$$

W.U Give the Cartesian coordinates of vector field

$$\vec{H} = 20\hat{a}_\rho - 10\hat{a}_\phi + 3\hat{a}_z \text{ at pt } (x=5, y=2, z=-1)$$

$$\begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} H_\rho \\ H_\phi \\ H_z \end{bmatrix}$$

$$\phi = \tan^{-1}(y/x) = \tan^{-1}(2/5) = 21.8^\circ$$

$$H_x = H_\rho \cos \phi - H_\phi \sin \phi$$

$$= 20 \cos 21.8 + 10 \sin 21.8 = 22.282$$

$$H_y = H_\rho \sin \phi + H_\phi \cos \phi = -1.856$$

$$H_z = H_z = 3$$

Transformation of vectors from Cartesian to Spherical :-

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$$

We transform it into spherical system. Component

g $\vec{A}$ in $\vec{a}_e$ direction

$$A_T = \vec{A} \cdot \vec{a}_e = [A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z] \cdot \vec{a}_e$$

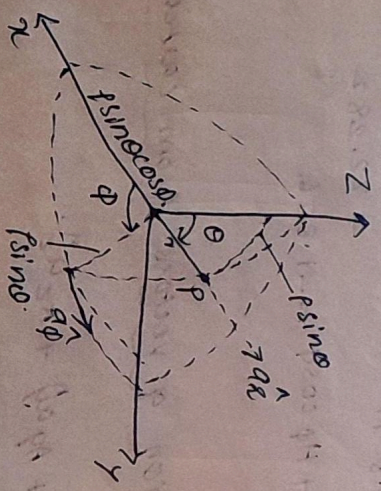
$$= A_x (\hat{a}_x \cdot \vec{a}_e) + A_y (\hat{a}_y \cdot \vec{a}_e) + A_z (\hat{a}_z \cdot \vec{a}_e)$$

$$A_\theta = \vec{A} \cdot \vec{a}_\theta = [A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z] \cdot \vec{a}_\theta$$

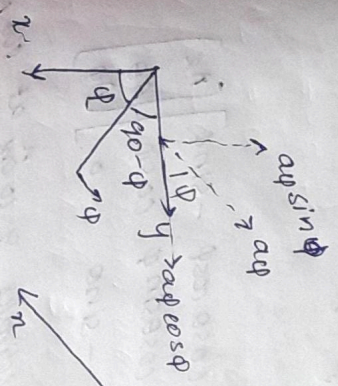
$$= A_x \hat{a}_x \cdot \vec{a}_\theta + A_y \hat{a}_y \cdot \vec{a}_\theta + A_z \hat{a}_z \cdot \vec{a}_\theta$$

$$A_\phi = \vec{A} \cdot \vec{a}_\phi = A_x \hat{a}_x \cdot \vec{a}_\phi + A_y \hat{a}_y \cdot \vec{a}_\phi + A_z \hat{a}_z \cdot \vec{a}_\phi$$

	$\hat{a}_x$	$\hat{a}_\theta$	$\hat{a}_\phi$
$\hat{a}_x$	$\sin\theta \cos\phi$	$\cos\theta \cos\phi$	$-\sin\phi$
$\hat{a}_y$	$\sin\theta \sin\phi$	$\cos\theta \sin\phi$	$\cos\phi$
$\hat{a}_z$	$\cos\theta$	$-\sin\theta$	0



ϕ



root quadrants	$\hat{a}_x$	$\hat{a}_\theta$	$\hat{a}_\phi$
$\hat{a}_x$	$\sin\theta \cos\phi$	$\cos\theta \cos\phi$	$-\sin\phi$
$\hat{a}_y$	$\sin\theta \sin\phi$	$\cos\theta \sin\phi$	$\cos\phi$
$\hat{a}_z$	$\cos\theta$	$-\sin\theta$	0

Cartesian to spherical

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta \cos\phi & \sin\theta \sin\phi & \cos\theta \\ \cos\theta \cos\phi & \cos\theta \sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Relation of vectors from spherical to Cartesian.

$$\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi$$

$$A_x = (\hat{r}) \cdot \hat{a}_x = A_x \hat{a}_x \cdot \hat{a}_x + A_y \hat{a}_y \cdot \hat{a}_x + A_z \hat{a}_z \cdot \hat{a}_x$$

$$A_y = A_x \hat{a}_x \cdot \hat{a}_y + A_y \hat{a}_y \cdot \hat{a}_y + A_z \hat{a}_z \cdot \hat{a}_y$$

$$A_z = (\hat{r}) \cdot \hat{a}_z$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Obtain the spherical coordinates of 10 cm at the point $P(-3, 2, 4)$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Given vector $\bar{A} = 10 \hat{a}_y$

$$F_x = \bar{F} \cdot \hat{a}_x = 10 \hat{a}_y \cdot \hat{a}_x$$

$$= 10 \sin \theta \cos \phi$$

$$\phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \left(\frac{2}{-3} \right) = -38.42^\circ$$

$$\phi = 180^\circ - 38.42^\circ = 141.58^\circ, \cos \phi = -0.832$$

$$\sin \phi = 0.554$$

$$\theta = \cos^{-1} z/r =$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{9 + 4 + 16} = 5.385$$

$$\theta = \frac{4}{5.38} = 42.03^\circ$$

$$\cos \theta = 0.742, \sin \theta = 0.669$$

$$F_x = 10 \times 0.669 \times 0.554$$

$$= -5.57$$

$$F_\theta = 10 \hat{a}_y \cdot \hat{a}_\theta = 10 \cos \theta \cos \phi$$

$$= 10 \times 0.742 \times (-0.832)$$

$$= -6.18$$

$$F_\phi = -10 \times 0.554 = -5.54$$

$$\bar{F} = -5.57 \hat{a}_x - 6.18 \hat{a}_\theta - 5.54 \hat{a}_\phi$$

Express $\bar{B} = x^2 \hat{a}_x + \sin \theta \hat{a}_y$ in the cartesian coordinates

Obtain $\bar{B}$ at $P(1, 2, 3)$

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix}$$

$$B_x = B_x \sin \theta \cos \phi + \sin \theta \sin \phi$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{14} = 3.74$$

$$\cos \theta = z/r = \frac{3}{3.74}$$

$$\theta = 36.7^\circ$$

$$\tan \phi = y/x = 2$$

$$\phi = 63.43^\circ$$

$$B_x = (3-74)^2 \times \sin 36.7^\circ \times \cos 63.43^\circ$$

$$= 3.904$$

$$B_y = r^2 \sin \theta \sin \phi + \cos \phi \sin \theta$$

$$= 7.743$$

$$B_z = r^2 \cos \theta = 11.21$$

Transformation of vectors from spherical to cylindrical

$$\bar{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi$$

$$\bar{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_\phi \hat{a}_\phi$$

$$A_r = A_r \hat{a}_r \cdot \hat{a}_r + A_\theta \hat{a}_\theta \cdot \hat{a}_r + A_\phi \hat{a}_\phi \cdot \hat{a}_r$$

$$A_\theta = A_r \hat{a}_r \cdot \hat{a}_\theta + A_\theta \hat{a}_\theta \cdot \hat{a}_\theta + A_\phi \hat{a}_\phi \cdot \hat{a}_\theta$$

$$A_z = A_r \hat{a}_r \cdot \hat{a}_z + A_\theta \hat{a}_\theta \cdot \hat{a}_z + A_\phi \hat{a}_\phi \cdot \hat{a}_z$$

	$\hat{a}_r$	$\hat{a}_\theta$	$\hat{a}_\phi$
$\hat{a}_r$	$\sin \theta \cos \theta$	0	0
$\hat{a}_\theta$	0	1	0
$\hat{a}_z$	$\cos \theta$	$-\sin \theta$	0

$$\begin{bmatrix} A_r \\ A_\theta \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \theta & 0 & 0 \\ 0 & 1 & 0 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_z \end{bmatrix}$$

Transformation of vectors from cylindrical to spherical

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_z \end{bmatrix}$$

Del operator :-

∇ is called vector differential operator

Cartesian coordinate system

$$\nabla = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z$$

Cylindrical coordinate system

$$\nabla = \frac{\partial}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} \hat{a}_\phi + \frac{\partial}{\partial z} \hat{a}_z$$

Spherical coordinate system

$$\nabla = \frac{\partial}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \hat{a}_\phi$$

Gradient of a scalar field $[\nabla \psi]$

1. Gradient of a scalar $\bar{F}$, $\nabla \bar{F} \rightarrow$ vector
2. Divergence of vector $\bar{F}$, $\nabla \cdot \bar{F} \rightarrow$ scalar
3. Curl of vector $\bar{F} \rightarrow \nabla \times \bar{F} \rightarrow$ vector

gradient of a scalar field V is a vector that presents both the magnitude & the direction of maximum rate of change of V .

For cartesian coordinate system
grad $V = \nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$

For cylindrical coordinate system

$$\nabla V = \frac{\partial V}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{\phi} + \frac{\partial V}{\partial z} \hat{z}$$

For spherical coordinate

$$\nabla V = \frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{\phi}$$

Properties of gradient

1) Magnitude of ∇V = maximum rate of change in V per unit distance

2) ∇V points in the direction of maximum rate of change in V .

3) If $\vec{A} = \nabla V$, then V is said to be scalar potential of $\vec{A}$

4) The component of ∇V in the direction of a unit vector $\hat{a}_n$ is $\nabla V \cdot \hat{a}_n$ and is called directional derivative of V along $\hat{a}_n$.

Find the gradient of following scalar fields

a) $V = e^{-z} \sin 2x \cos y$

b) $V = \rho^2 z \cos 2\phi$

c) $W = 10r^4 \sin \theta \cos \phi$

d) $F = 6\rho^4 z^3 \sin \phi$

a) $\nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$

$$= e^{-z} \cos y \cdot 2 \cos 2x \hat{a}_x + e^{-z} \sin 2x \sin y \hat{a}_y$$

$$- e^{-z} \sin 2x \sin y \hat{a}_z$$

$$= 2e^{-z} \cos y \cos 2x \hat{a}_x + e^{-z} \sin 2x \sin y \hat{a}_y$$

$$- e^{-z} \sin 2x \sin y \hat{a}_z$$

b) $\nabla V = \frac{\partial V}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{\phi} + \frac{\partial V}{\partial z} \hat{z}$

$$= 2\rho z \cos 2\phi \hat{\rho} + \frac{1}{\rho} \rho^2 z \sin 2\phi \hat{\phi} +$$

$$\rho^2 \cos 2\phi \hat{z}$$

$$= 2\rho z \cos 2\phi \hat{\rho} - 2\rho z \sin 2\phi \hat{\phi} +$$

$$\rho^2 \cos 2\phi \hat{z}$$

c) $\nabla W = \frac{\partial W}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial W}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial W}{\partial \phi} \hat{\phi}$

$$= 40r^3 \sin \theta \cos \phi \hat{r} + \frac{1}{r} 10r^4 \cos \theta \cos \phi \hat{\theta} +$$

$$\frac{1}{r \sin \theta} 10r^4 \sin \theta \sin \phi \hat{\phi}$$

$$= 40r^3 \sin \theta \cos \phi \hat{r} + 10r^3 \cos \theta \cos \phi \hat{\theta}$$

$$- 10r^3 \sin \phi \hat{\phi}$$

$$\begin{aligned}
 d) \nabla f &= \frac{\partial f}{\partial \rho} \hat{\rho} + \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z} \\
 &= 20\rho^2 z^2 \sin \phi \hat{\rho} + 5\rho^4 z^3 \cos \phi \hat{\phi} + 5\rho^4 z^2 \sin \phi \hat{z} \\
 &= 20\rho^2 z^2 \sin \phi \hat{\rho} + 5\rho^4 z^3 \cos \phi \hat{\phi} + 5\rho^4 z^2 \sin \phi \hat{z}
 \end{aligned}$$

Find the gradient of scalar function $V = \rho^2 \sin \phi$ in cylindrical coordinate $\sin \phi$ & directional derivative in the direction of vector $\vec{r} = \hat{\rho} + \hat{\phi}$ at the point $[2, \pi/4, 0]$

$$V = \rho^2 \sin \phi$$

$$\begin{aligned}
 \nabla V &= \frac{\partial V}{\partial \rho} \hat{\rho} + \frac{\partial V}{\partial \phi} \hat{\phi} + \frac{\partial V}{\partial z} \hat{z} \\
 &= 2\rho \sin \phi \hat{\rho} + \rho^2 \cos \phi \hat{\phi} + 0
 \end{aligned}$$

$$\text{At } P[2, \pi/4, 0]$$

$$\nabla V = 4 \sin 2 \times \frac{\pi}{4} \hat{\rho} + 2 \times \cos 2 \times \frac{\pi}{4} \hat{\phi} \cdot 2$$

$$= 4 \hat{\phi}$$

Directional derivative..

$$= \nabla V \cdot \hat{a}_0$$

$$= 4 \hat{\phi} \cdot \left[\frac{\hat{\rho} + \hat{\phi}}{\sqrt{2}} \right] = \frac{4}{\sqrt{2}} = \underline{\underline{2\sqrt{2}}}$$

Given $\phi = xy + yz + xz$. Find the gradient of ϕ at pt $(1, 2, 3)$ and the directional derivative of ϕ at the same point in the direction of point $(3, 4, 4)$.

$$\nabla \phi = \frac{\partial \phi}{\partial x} \hat{a}_x + \frac{\partial \phi}{\partial y} \hat{a}_y + \frac{\partial \phi}{\partial z} \hat{a}_z$$

$$= (y+z) \hat{a}_x + (x+z) \hat{a}_y + (x+y) \hat{a}_z$$

$$\nabla \phi \text{ at } (1, 2, 3)$$

$$= 5 \hat{a}_x + 4 \hat{a}_y + 3 \hat{a}_z$$

vector joining pts $(1, 2, 3)$ and $(3, 4, 4)$

$$\vec{r} = 2\hat{a}_x + 2\hat{a}_y + \hat{a}_z$$

Directional derivative

$$\begin{aligned}
 &= \nabla \phi \cdot \hat{a}_n \\
 &= [5\hat{a}_x + 4\hat{a}_y + 3\hat{a}_z] \cdot \frac{[2\hat{a}_x + 2\hat{a}_y + \hat{a}_z]}{\sqrt{4+4+1}}
 \end{aligned}$$

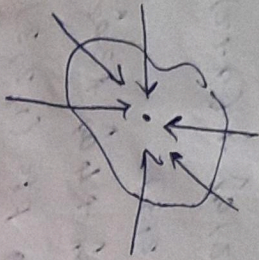
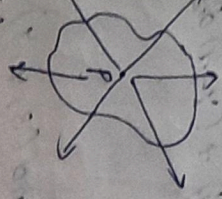
$$= \frac{10+8+3}{3} = \underline{\underline{7}}$$

Divergence:-

Divergence of a vector quantity $\vec{A}$ is the outward flow of vector qty per unit volume from a closed surface surrounding the volume as the volume tends to zero.

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta V \rightarrow 0} \frac{\oint \vec{A} \cdot d\vec{s}}{\Delta V}$$

At any point, the field lines are diverging/coming out from a small volume surrounding the point. (Fig a)



(b)

The field lines are converging into a small volume the divergence of vector is -ve (Fig b)
The rate at which field lines are entering a small volume surrounding the pt = the rate at which they are leaving the same volume then the divergence of the vector is zero (Fig c)

$$\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$$

$$\nabla = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z$$

$$\nabla \cdot \vec{r} = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y + \frac{\partial}{\partial z}z$$

significance of a vector qty is as scalar.

Cylindrical coordinate sim.

$$\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

Spherical coordinate sim.

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} A_\phi$$

Properties of divergence.

Vector Field

1) Divergence produces a scalar field
divergence of scalar field $\rightarrow$ vector field.

2) Distributed law

$$\nabla \cdot (A+B) = \nabla \cdot A + \nabla \cdot B$$

3) Any vector whose divergence is zero is called solenoidal vector. $\nabla \cdot \vec{A} = 0$.

? Determine the divergence of the vector field.

a) $\vec{P} = x^2 y z \hat{a}_x + x z \hat{a}_z$

b) $\vec{Q} = \rho \sin \phi \hat{a}_\rho + \rho^2 z \hat{a}_\phi + z \cos \phi \hat{a}_z$

c) $\vec{T} = \frac{1}{r^2} \cos \theta \hat{a}_r + r \sin \theta \cos \phi \hat{a}_\theta + \cos \theta \hat{a}_\phi$

a) $\vec{P} = x^2 y z \hat{a}_x + x z \hat{a}_z$

$$\nabla \cdot \vec{P} = \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (x z)$$

$$= 2 x y z + x$$

$$b) \nabla \cdot \mathbf{a} = \rho \sin \phi \frac{\partial}{\partial \rho} \hat{\rho} + \rho^2 z \frac{\partial}{\partial \phi} \hat{\phi} + z \cos \phi \frac{\partial}{\partial z} \hat{z}$$

$$\nabla \cdot \mathbf{a} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \rho (\rho \sin \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} \rho^2 z + \frac{\partial}{\partial z} z \cos \phi$$

$$= \frac{1}{\rho} \cdot 2\rho \sin \phi + \frac{1}{\rho} \times 0 + \cos \phi$$

$$= 2 \sin \phi + \cos \phi$$

$$c) \nabla \cdot \mathbf{T} = \frac{1}{r^2} \cos \theta \frac{\partial}{\partial r} (r \sin \theta \cos \phi) \hat{r} + \cos \theta \frac{\partial}{\partial \theta} \hat{\theta}$$

$$\nabla \cdot \mathbf{T} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot \frac{1}{r^2} \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin \theta \cos \phi)$$

$$+ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \cos \phi$$

$$= \frac{1}{r^2} \cdot 0 + \frac{1}{r \sin \theta} \cdot 2 \sin \theta \cos \theta \cos \phi + \frac{1}{r \sin \theta} \cdot 0$$

$$= \underline{\underline{2 \cos \theta \cos \phi}}$$

Find the constant so that $\nabla \cdot (\mathbf{x} + 3y)\hat{\mathbf{a}}_x + (y - 3x)\hat{\mathbf{a}}_y + (x + yz)\hat{\mathbf{a}}_z$ is solenoidal

∇ is solenoidal means $\nabla \cdot \mathbf{V} = 0$

$$\frac{\partial}{\partial x} (x + 3y) + \frac{\partial}{\partial y} (y - 3x) + \frac{\partial}{\partial z} (x + yz) = 0$$

$$1 + 1 + a = 0$$

$$a = -2$$

2 Find the divergence of $\mathbf{A}$ at pt $P(5, \eta_2, 1)$

where $\mathbf{A} = r z \sin \phi \hat{\mathbf{a}}_r + 3r z^2 \cos \phi \hat{\mathbf{a}}_\phi$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} r A_r + \frac{1}{r} \frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

$$= \frac{1}{r} \frac{\partial}{\partial r} r^2 z \sin \phi + \frac{1}{r} \frac{\partial}{\partial \phi} (3r z^2 \cos \phi) + 0$$

$$= \frac{1}{r} \cdot 2r z \sin \phi + \frac{1}{r} \cdot 3r z^2 \cdot (-\sin \phi)$$

$$= 2 z \sin \phi - 3 z^2 \sin \phi$$

$$\text{At pt } P(5, \eta_2, 1)$$

$$\nabla \cdot \mathbf{A} = 2 \times 1 \times \sin(\eta_2) - 3 \times 1 \times \sin \eta_2$$

$$= 2 - 3 = \underline{\underline{-1}}$$

Ques

→ curl provides the measure of amt of rotation or angular momentum of a vector around the pt.

→ The curl of a vector field is the measure of the maximum circulation of the vector field around the boundary of an infinitesimal area per unit area as area tends to zero.

$$\text{curl } \mathbf{A} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{A} \cdot d\mathbf{l} \hat{\mathbf{a}}_n$$

where ΔS is the area bounded by curve C

indicates the direction along which maximum variation occurs.

In vector analysis

$$\text{curl } \vec{A} = \nabla \times \vec{A}$$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}; \text{ cartesian s/m}$$

$$\nabla \times \vec{A} = \frac{1}{r} \begin{vmatrix} \hat{a}_r & r \hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix}, \text{ cylindrical}$$

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{a}_r & r \hat{a}_\theta & r \sin \theta \hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

for spherical coord.

$$\nabla \times \vec{A} = 0$$

, then A is called irrotational.

Find the curl of the following vector fields

a) $P = x^2 y z \hat{a}_x + x z \hat{a}_z$

b) $H = 2 y a_x + (z^2 - x^2) a_y + 3 y a_z$

c) $B = 2 \rho \cos \phi \hat{a}_\rho - 4 \rho \sin \phi \hat{a}_\phi + 3 \hat{a}_z$

a) $\text{curl } P = \nabla \times P = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y z & 0 & x z \end{vmatrix}$

$$= \hat{a}_x \left[\frac{\partial}{\partial y} (x z) - 0 \right] - \hat{a}_y \left[\frac{\partial}{\partial x} (x z) - \frac{\partial}{\partial z} (x^2 y z) \right] + \hat{a}_z \left[0 - \frac{\partial}{\partial y} (x^2 y z) \right]$$

$$= - \left[z - x^2 y \right] \hat{a}_y - x^2 z \hat{a}_z = (x^2 y z) \hat{a}_y - x^2 z \hat{a}_z$$

b) $\text{curl } H = \nabla \times H = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2 y & z^2 - x^2 & 3 y \end{vmatrix}$

$$= \hat{a}_x \left(\frac{\partial}{\partial y} 3 y - \frac{\partial}{\partial z} (z^2 - x^2) \right) - \hat{a}_y \left[\frac{\partial}{\partial x} (z^2 - x^2) - \frac{\partial}{\partial z} 2 y \right]$$

$$= a_1' [3-2z] - 0 + a_2' [-2x-2]$$

$$= [3-2z] a_1' - [2x+2] a_2'$$

$$\text{curl } \bar{B} = \nabla \times \bar{B} = \frac{1}{\rho} \begin{vmatrix} \hat{a}_\rho & \rho \hat{a}_\phi & a_z' \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$= \frac{1}{\rho} \begin{vmatrix} \hat{a}_\rho & \rho \hat{a}_\phi & a_z' \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 2\rho \cos \phi & -4\rho^2 \sin \phi & 3 \end{vmatrix}$$

$$= \frac{1}{\rho} \left[\hat{a}_\rho \left(\frac{\partial}{\partial \phi} \cdot 3 - \frac{\partial}{\partial z} 4\rho^2 \sin \phi \right) - \rho \hat{a}_\phi \left[\frac{\partial}{\partial \rho} 3 - \frac{\partial}{\partial z} (2 \cos \phi) \right] \right.$$

$$\left. + \hat{a}_z \left[\frac{\partial}{\partial \rho} (-4\rho^2 \sin \phi) - \frac{\partial}{\partial \phi} (2 \cos \phi) \right] \right]$$

$$= \frac{1}{\rho} \left[\hat{a}_\rho (0) - \rho \hat{a}_\phi [0] + \hat{a}_z [-8\rho \sin \phi + 2 \sin \phi] \right]$$

$$= [2 \sin \phi - 8 \sin \phi] \hat{a}_z = -6 \sin \phi \hat{a}_z$$

? A vector $\bar{V}$ is irrotational, find the constants a, b, c . $\bar{V} = (x+2y+az) \hat{a}_x + (bx-3y-z) \hat{a}_y + (4x+cy+2z) \hat{a}_z$

$$\nabla \times \bar{V} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix}$$

$$= a_1' \left[\frac{\partial}{\partial y} (4x+cy+2z) - \frac{\partial}{\partial z} (bx-3y-z) \right] - \hat{a}_y \left[\frac{\partial}{\partial x} (4x+cy+2z) - \frac{\partial}{\partial z} (x+2y+az) \right]$$

$$= a_1' [c+1] - \hat{a}_y [4-a] + a_z' [b-2]$$

$$\nabla \times \bar{V} = 0$$

$$c+1=0 \Rightarrow c=-1$$

$$4-a=0 \Rightarrow a=4$$

$$b-2=0 \Rightarrow b=2$$

W.V

vector field is given by $\bar{A} = yz \hat{a}_x + xz \hat{a}_y + xy \hat{a}_z$
S.T it is irrotational.

$$\nabla \times \bar{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix}$$

$$= a_1' \left(\frac{\partial}{\partial y} (xy) - \frac{\partial}{\partial z} (xz) \right) - \hat{a}_y \left(\frac{\partial}{\partial x} (xy) - \frac{\partial}{\partial z} (yz) \right) + \hat{a}_z \left(\frac{\partial}{\partial x} (xz) - \frac{\partial}{\partial y} (yz) \right)$$

$$= \hat{a}_x [x - x] - \hat{a}_y [y - y] + \hat{a}_z [z - z] = 0$$

Divergence theorem / Gauss's divergence theorem

The theorem states that total outward flux of vector field $\vec{A}$ through the closed surface S is the same as the vol. integral of divergence of that vector:

$$\int_V \nabla \cdot \vec{A} \, dv = \oint_S \vec{A} \cdot d\vec{s}$$

Proof (cartesian sim)

Divergence of any vector

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Volume integral on both sides

$$\int_V \nabla \cdot \vec{A} \, dv = \int_V \left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right] dv$$

$$= \int_V \left[\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right] dv$$

$$= \int_V \frac{\partial A_x}{\partial x} dv = \int \int \left[\frac{\partial A_x}{\partial x} \right] dy dz$$

$$\text{Let } \int_V \frac{\partial A_x}{\partial x} dv = A_x|_{x=0} - A_x|_{x=L} = A_x$$

$$\text{So } \int_V \frac{\partial A_x}{\partial x} dv = \int_0^L A_x dy dz = \int_0^L A_x dx$$

where $dx = dy dz$ is the x component of surface area $d\vec{s}$.

Similarly

$$B = \int_V \frac{\partial A_y}{\partial y} dv = \int_0^L A_y dx dz$$

$$= \int_0^L A_y dy$$

$$C = \int_V \frac{\partial A_z}{\partial z} dv = \int_0^L A_z dx dy$$

Subst in (1)

$$\int_V \nabla \cdot \vec{A} \, dv = \int_0^L A_x dx + \int_0^L A_y dy + \int_0^L A_z dz$$

$$= \oint_S \vec{A} \cdot d\vec{s}$$

$$\int_V \nabla \cdot \vec{A} \, dv = \oint_S \vec{A} \cdot d\vec{s}$$

Divergence theorem applies to time varying as well as static fields. This theorem is used to convert volume integral of a divergence of a vector into a closed surface integral.

ing divergence theorem evaluate $\oint \vec{F} \cdot d\vec{A}$ for
 $\vec{F} = 2xy\hat{x} + y^2\hat{y} + 4yz\hat{z}$ and S is
 surface of the cube bounded by $x=0, x=1,$
 $y=0, y=1, z=0, z=1$.

divergence theorem

$$\oint_S \vec{F} \cdot d\vec{A} = \iiint_V \nabla \cdot \vec{F} \, dV$$

$$\vec{F} = \frac{\partial}{\partial x} (2xy) + \frac{\partial}{\partial y} (y^2) + \frac{\partial}{\partial z} (4yz)$$

$$= \frac{\partial}{\partial x} (2xy) + \frac{\partial}{\partial y} (y^2) + \frac{\partial}{\partial z} (4yz)$$

$$= 2y + 2y + 4y = 8y$$

$$\nabla \cdot \vec{F} \, dV = \iiint_V 8y \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^1 \int_0^1 8y \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^1 [8y x]_0^1 \, dy \, dz$$

$$= \int_0^1 \int_0^1 8y \, dy \, dz$$

$$= \int_0^1 [8 \frac{y^2}{2}]_0^1 \, dz$$

$$= \int_0^1 4 \, dz = 4[z]_0^1 = 4$$

$$\oint \vec{F} \cdot d\vec{A} = 4$$

with volume of divergence theorem considering
 the fluid $\vec{D} = 2xy\hat{x} + x^2y\hat{y} + \hat{z}$ and the surface
 is a parallelepiped formed by planes $x=0, y=1,$
 $y=0, z=2, z=0, z=3$.

$$\oint \vec{D} \cdot d\vec{A} = \iiint_V \nabla \cdot \vec{D} \, dV$$

RHS:

$$\nabla \cdot \vec{D} = \frac{\partial}{\partial x} (2xy) + \frac{\partial}{\partial y} (x^2y) + \frac{\partial}{\partial z} (1)$$

$$= \frac{\partial}{\partial x} (2xy) + \frac{\partial}{\partial y} (x^2y) + \frac{\partial}{\partial z} (1)$$

$$= 2y + 0 = 2y$$

$$\iiint_V \nabla \cdot \vec{D} \, dV = \int_0^1 \int_0^2 \int_0^3 2y \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^2 \int_0^3 2y \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^2 [2y x]_0^3 \, dy \, dz$$

$$= \int_0^1 \int_0^2 6y \, dy \, dz = \int_0^1 [3y^2]_0^2 \, dz$$

$$= \int_0^1 3 \cdot [4-0] \, dz = \int_0^1 12 \, dz = 12[z]_0^1 = 12$$

L.H.S:

$$\oint \vec{D} \cdot d\vec{A} = \int \vec{D} \cdot \hat{n} \, dA + \int \vec{D} \cdot (-\hat{n}) \, dA + \int \vec{D} \cdot \hat{n} \, dA + \int \vec{D} \cdot (-\hat{n}) \, dA$$

$$= \int \vec{D} \cdot \hat{n} \, dA + \int \vec{D} \cdot (-\hat{n}) \, dA + \int \vec{D} \cdot \hat{n} \, dA + \int \vec{D} \cdot (-\hat{n}) \, dA$$

$$\int_0^3 \int_0^2 \int_0^2 [2xy \hat{a}_x + x^2 \hat{a}_y] \cdot \hat{a}_n \, dy \, dz$$

$$= \int_0^3 \int_0^2 2xy \, dy \, dz = \int_0^3 [xy^2]_0^2 \, dz$$

$$= \int_0^3 x [4 - 0] \, dz = 4x [z]_0^3 = 12x \quad (x=1)$$

$$\int \int \nabla \cdot (-\hat{a}_n) \, dy \, dz$$

$$= \int_0^3 \int_0^2 (2xy \hat{a}_x + x^2 \hat{a}_y) \cdot (-\hat{a}_n) \, dy \, dz$$

$$= \int_0^3 \int_0^2 -2xy \, dy \, dz = \int_0^3 -x [y^2]_0^2 \, dz$$

$$= \int_0^3 -x \left(\frac{4}{2}\right) \, dz = -4x [z]_0^3$$

$$= -12x = 0 \quad [x=0]$$

$$\int \int \nabla \cdot \hat{a}_y \, dx \, dz = \int_0^3 \int_0^1 (2xy \hat{a}_x + x^2 \hat{a}_y) \cdot \hat{a}_y \, dx \, dz$$

$$= \int_0^3 \int_0^1 x^2 \, dx \, dz = \int_0^3 \left[\frac{x^3}{3}\right]_0^1 \, dz$$

$$= \int_0^3 \frac{1}{3} \, dz = \frac{1}{3} [z]_0^3 = 1$$

$$\int \int \nabla \cdot \hat{a}_y \, dx \, dz = \int_0^3 \int_0^1 (2xy \hat{a}_x + x^2 \hat{a}_y) \cdot \hat{a}_y \, dx \, dz$$

$$= \int_0^3 \int_0^1 -x^2 \, dx \, dz = \int_0^3 \left[-\frac{x^3}{3}\right]_0^1 \, dz$$

$$= \int_0^3 -\frac{1}{3} \, dz = -\frac{1}{3} [z]_0^3 = -1$$

$$\int_0^3 \int_0^2 \nabla \cdot \hat{a}_z \, dx \, dy = \int_0^3 \int_0^2 (2xy \hat{a}_x + x^2 \hat{a}_y) \cdot \hat{a}_z \, dx \, dy$$

$$= 0$$

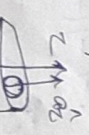
Z plane constant since D lies in Z plane any x & y is arbitrary

$$\int \int \nabla \cdot \hat{a}_z \, dx \, dy = 0$$

$$LHS = \int \int \nabla \cdot \hat{n} \, ds = 12 + 0 + 1 + -1 + 0 + 0 = 12$$

LHS = RHS - Divergence theorem is verified.

Given $\vec{A} = 30e^{-\rho} \hat{a}_\rho - 2z \hat{a}_z$ in cylindrical coordinates. Evaluate both sides of divergence theorem for volume enclosed by $\rho=2, z=0$ & $z=5$.



$$\iiint_V \nabla \cdot \vec{A} \, dv = \oiint_S \vec{A} \cdot d\vec{s}$$

LHS

$$\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \cdot 30e^{-\rho}) + 0 + \frac{\partial}{\partial z} (-2z)$$

$$= \frac{1}{\rho} [\rho \cdot 30e^{-\rho} - 1 + 30e^{-\rho} \cdot 1] - 2$$

$$= -\frac{f}{\rho 30e^{-f} + 30e^{-f} - 2f}$$

$$= \frac{30e^{-f} - 30pe^{-f} - 2f}{f}$$

$$\iiint_V \nabla \cdot A \, dV = \iiint_V \frac{30e^{-f} - 30pe^{-f} - 2f}{f} \cdot f \, d\rho \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} \int_0^2 (30e^{-f} - 30pe^{-f} - 2f) \, d\rho \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} [-30e^{-f} - 30fpe^{-f} - \int_0^2 f \, d\rho] \, d\phi \, dz$$

$$-2\left[\frac{f^2}{2}\right]_0^2 \, d\phi \, dz$$

$$\int \rho e^{-f} \, d\rho = \rho \int e^{-f} \, d\rho - \int (1 \int e^{-f} \, d\rho) \, d\rho$$

$$= \rho e^{-f} \cdot -1 - \int (-e^{-f}) \, d\rho$$

$$= -\rho e^{-f} + e^{-f} \rho - 1$$

$$= -\rho e^{-f} - e^{-f}$$

$$\iiint_V \nabla \cdot A \, dV = \int_0^5 \int_0^{2\pi} [-30e^{-f} - 30fpe^{-f} - e^{-f} - 2\frac{f^2}{2}] \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} [-30e^{-f} + 30pe^{-f} + 30e^{-f} - f^2] \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} 30pe^{-f} - f^2 \, d\phi \, dz$$

$$= \int_0^5 \int_0^{2\pi} (30 \times 2 e^{-2} - 4) \, d\phi \, dz = \int_0^5 \int_0^{2\pi} 4 \cdot 12 \, d\phi \, dz$$

$$= \int_0^5 4 \cdot 12 \, dz = \int_0^5 4 \cdot 12 \cdot 2\pi \, dz$$

$$= 4 \cdot 12 \cdot 2\pi \cdot 5 =$$

$$= \underline{\underline{129.48}}$$

RHS

$$\oint A \cdot ds = \int_1 A \cdot ds + \int_2 A \cdot ds + \int_3 A \cdot ds$$

$$\textcircled{1} \int A \cdot ds = \int_0^{2\pi} \int_0^2 [30e^{-f} \hat{\rho} - 2Z \hat{a}_z] \cdot \rho \, d\rho \, d\phi \, \hat{a}_z$$

$$= \int_0^{2\pi} \int_0^2 -2Z \rho \, d\rho \, d\phi = \int_0^{2\pi} [-2Z \frac{\rho^2}{2}]_0^2 \, d\phi$$

$$= \int_0^{2\pi} -2Z \cdot \frac{4}{2} \, d\phi = -4Z \phi \Big|_0^{2\pi} = -8\pi Z \Big|_0^5$$

$$= \underline{\underline{-40\pi}}$$

$$\textcircled{2} \int A \cdot ds = \int_0^{2\pi} \int_0^2 (30e^{-f} \hat{\rho} - 2Z \hat{a}_z) \cdot \rho \, d\rho \, d\phi \, \hat{a}_z$$

$$= \int_0^{2\pi} \int_0^2 +2Z \rho \, d\rho \, d\phi = \int_0^{2\pi} 2Z [\frac{\rho^2}{2}]_0^2 \, d\phi$$

$$= \int_0^{2\pi} 2 \cdot 4 \, d\phi = 4Z \cdot 2\pi$$

$$= 8\pi Z$$

$$= 0; Z=0$$

$$\iint_A ds = \iint [30e^{-\rho} \alpha_\rho^A - 2z\alpha_z^A] \cdot \rho d\rho dz \alpha_\rho^A$$

③

$$= \int_0^5 \int_0^{2\pi} 30e^{-\rho} \cdot \rho d\rho dz$$

$$= \int_0^5 30e^{-\rho} \cdot \rho \cdot \int_0^{2\pi} dz$$

$$= \int_0^5 30e^{-\rho} \cdot \rho \cdot 2\pi dz$$

$$= 60\pi e^{-\rho} \cdot \rho \cdot z \Big|_0^5 = 300\pi \int_0^5 \rho e^{-\rho}$$

$$\int_0^5 \rho e^{-\rho} d\rho = 2$$

$$= 81 \cdot 2\pi$$

$$\oint_A ds = -40\pi + 0 + 81 \cdot 2\pi$$

$$= 129.43$$

$$\underline{\underline{LHS = RHS}}$$

1. Verify divergence theorem for the vector $A = 4x \hat{a}_x - 2y^2 \hat{a}_y + z^2 \hat{a}_z$ taken over the cube bounded by $x=0, x=1, y=0, y=1$, and $z=0, z=1$

Divergence theorem is given by,

$$\iiint_V \nabla \cdot A \, dv = \iint_S A \cdot ds$$

LHS :

$$\nabla \cdot A = \frac{\partial}{\partial x} A_x + \frac{\partial}{\partial y} A_y + \frac{\partial}{\partial z} A_z$$

$$= \frac{\partial}{\partial x} 4x + \frac{\partial}{\partial y} (-2y^2) + \frac{\partial}{\partial z} z^2$$

$$= 4 - 4y + 2z$$

$$\iiint_V \nabla \cdot A \, dv = \int_0^1 \int_0^1 \int_0^1 4 - 4y + 2z \, dx \, dy \, dz$$

$$= \int_0^1 \int_0^1 [4x - 4xy + 2zx]_0^1 \, dy \, dz$$

$$= \int_0^1 \int_0^1 4 - 4y + 2z \, dy \, dz$$

$$= \int_0^1 [4y - 4\frac{y^2}{2} + 2zy]_0^1 \, dz$$

$$= \int_0^1 [4 - 2 + 2z] \, dz$$

$$= \int_0^1 2 + 2z \, dz$$

$$= 22 + 2 \cdot \frac{2^2}{3} \int_0^1$$

$$= 22 + 1 = 23$$

$$\begin{aligned} \iint A \cdot d\mathbf{s} &= \iint A \cdot \hat{\mathbf{a}}_x \, dy \, dz + \iint A \cdot (-\hat{\mathbf{a}}_x) \, dy \, dz + \iint A \cdot \hat{\mathbf{a}}_y \, dx \, dz \\ &+ \iint A \cdot (-\hat{\mathbf{a}}_y) \, dx \, dz + \iint A \cdot \hat{\mathbf{a}}_z \, dx \, dy + \iint A \cdot (-\hat{\mathbf{a}}_z) \, dx \, dy \end{aligned}$$

$$\begin{aligned} \iint A \cdot \hat{\mathbf{a}}_x \, dy \, dz &= \int_0^1 \int_0^1 [4x \hat{\mathbf{a}}_x - 2y^2 \hat{\mathbf{a}}_y + z^2 \hat{\mathbf{a}}_z] \cdot \hat{\mathbf{a}}_x \, dy \, dz \\ &= \int_0^1 \int_0^1 4x \, dy \, dz \\ &= \int_0^1 [4x y]_0^1 \, dz \\ &= \int_0^1 4x \, dz \\ &= [4xz]_0^1 \\ &= 4x \\ &= 4 \quad [x=1] \end{aligned}$$

$$\begin{aligned} \iint A \cdot \hat{\mathbf{a}}_x \, dy \, dz &= \int_0^1 \int_0^1 [4x \hat{\mathbf{a}}_x - 2y^2 \hat{\mathbf{a}}_y + z^2 \hat{\mathbf{a}}_z] \cdot \hat{\mathbf{a}}_x \, dy \, dz \\ &= \int_0^1 \int_0^1 4x \, dy \, dz \\ &= \int_0^1 [4x y]_0^1 \, dz \end{aligned}$$

$$= \int_0^1 -4x \, dz$$

$$= -4xz \Big|_0^1$$

$$= -4x$$

$$= 0 \quad [x=0]$$

$$\iint A \cdot \hat{a}_y \, dx \, dz = \int_0^1 \int_0^1 [4x \hat{a}_x - 2y^2 \hat{a}_y + z^2 \hat{a}_z] \cdot \hat{a}_y \, dx \, dz$$

$$= \int_0^1 \int_0^1 -2y^2 \, dx \, dz$$

$$= \int_0^1 \left[-2y^2 x \right]_0^1 \, dz$$

$$= \int_0^1 -2y^2 \, dz$$

$$= -2y^2 z \Big|_0^1$$

$$= -2y^2$$

$$= -2 \quad [y=1]$$

$$\iint A \cdot \hat{a}_y \, dx \, dz = \int_0^1 \int_0^1 [4x \hat{a}_x - 2y^2 \hat{a}_y + z^2 \hat{a}_z] \cdot \hat{a}_y \, dx \, dz$$

$$= \int_0^1 \int_0^1 2y^2 \, dx \, dz$$

$$= 0 \quad ; [y=0]$$

$$\iint A \cdot \hat{a}_z dx dy = \int_0^1 \int_0^1 [x \hat{a}_x - y^2 \hat{a}_y + z^2 \hat{a}_z] \cdot \hat{a}_z dx dy$$

$$= \int_0^1 \int_0^1 z^2 dx dy$$

$$= \int_0^1 z^2 [x]_0^1 dy$$

$$= \int_0^1 z^2 dy$$

$$= z^2 [y]_0^1$$

$$= z^2 = 1 \quad ; \quad \{z=1\}$$

$$\iint A \cdot \hat{a}_z dx dy = \int_0^1 \int_0^1 [x \hat{a}_x - y^2 \hat{a}_y + z^2 \hat{a}_z] \cdot \hat{a}_z dx dy$$

$$= \int_0^1 \int_0^1 -z^2 dx dy$$

$$= -z^2 [x]_0^1 dy$$

$$= -z^2 [y]_0^1 = -z^2 = 0 \quad \{z=0\}$$

$$RHS = \oint A \cdot ds = 4 + 0 + 2 + 0 + 1 + 0$$

$$= 3$$

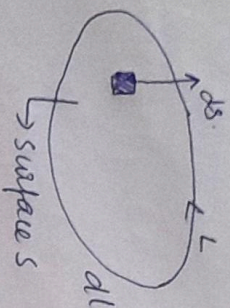
$$LHS = RHS$$

Hence divergence theorem is verified

Stokes Theorem

Stokes theorem relates the line integral to a surface integral. It states that the circulation of a vector field $\vec{H}$ around a closed path L is equal to the surface integral of the curl of $\vec{H}$ over the open surface S bounded by L .

$$\oint_L \vec{H} \cdot d\vec{l} = \iint_S (\nabla \times \vec{H}) \cdot d\vec{s} = \iint_S (\nabla \times \vec{H}) \cdot \vec{n} \, dA$$



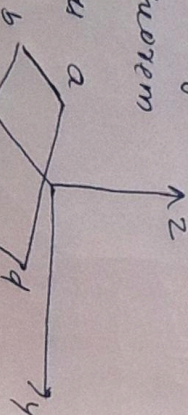
evaluate both sides of Stokes theorem for the vector field $\vec{H} = 6xy \hat{a}_x - 3y^2 \hat{a}_y$ A/m & the rectangular path around the region $2 \leq x \leq 5, -1 \leq y \leq 1, z=0$.

Ans. to Stokes theorem

$$\oint_L \vec{H} \cdot d\vec{l} = \iint_S (\nabla \times \vec{H}) \cdot d\vec{s}$$

Ans

$$\oint_L \vec{H} \cdot d\vec{l} = \int_a^b + \int_b^c + \int_c^d + \int_d^a \vec{H} \cdot d\vec{l}$$



Ans

$$\oint_L \vec{H} \cdot d\vec{l} = \int_a^b + \int_b^c + \int_c^d + \int_d^a \vec{H} \cdot d\vec{l}$$

$$= \int_a^b 6xy \, dx = \left[3y \frac{x^2}{2} \right]_2^5 = 3y [25 - 4] = 63y$$

for line ab value of $y = -1$

$$\int_a^b \vec{H} \cdot d\vec{l} = 63x - 1 = -43$$

$$\int_b^c \vec{H} \cdot d\vec{l} = \int_{-1}^1 (6xy \hat{a}_x - 3y^2 \hat{a}_y) \cdot dy \hat{a}_y$$

$$= \int_{-1}^1 -3y^2 \, dy = -\left[\frac{y^3}{3} \right]_{-1}^1 = -\frac{2}{3}$$

$$\int_c^d \vec{H} \cdot d\vec{l} = \int_2^5 (6xy \hat{a}_x - 3y^2 \hat{a}_y) \cdot dx \hat{a}_x$$

$$= \int_2^5 6xy \, dx = 6y \left[\frac{x^2}{2} \right]_2^5 = 3y [25 - 4] = 63y = -63 \times (-1) = 63$$

$$\int_d^a \vec{H} \cdot d\vec{l} = \int_{-1}^1 (6xy \hat{a}_x - 3y^2 \hat{a}_y) \cdot dy \hat{a}_y$$

$$= -\left[\frac{3y^3}{3} \right]_{-1}^1 = -[-1 - 1] = 2$$

$$\oint_L \vec{H} \cdot d\vec{l} = -63 - 2 - 63 + 2 = -126$$

$$\nabla \times \vec{H} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{vmatrix} = \hat{a}_x \left[0 - \frac{\partial}{\partial z}(-3y^2) \right] - \hat{a}_y \left[0 - \frac{\partial}{\partial z}(6xy) \right] + \hat{a}_z \left[\frac{\partial}{\partial x}(-3y^2) - \frac{\partial}{\partial y}(6xy) \right]$$

$$= \hat{a}_x (0) - \hat{a}_y (0) + \hat{a}_z (-6x)$$

$$= -6x \hat{a}_z$$

$$\iint_S \nabla \times \vec{H} \cdot d\vec{s} = \iint -6x \hat{a}_z \cdot x \hat{a}_y \hat{a}_z$$

$$= \int_{x=2}^5 \int_{y=-1}^1 -6x \, dx \, dy$$

$$= \int_{-1}^1 \left[-6 \frac{x^2}{2} \right]_2^5 dy = \int_{-1}^1 -3[25-4] dy$$

$$= -63y \Big|_{-1}^1 = \underline{\underline{-126}}$$

LHS = RHS